QP CODE: 24900272

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCZIM102

FUNDAMENTALS OF BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Fill In The Blanks
Answer all questions
Each question carries 1 mark

1	Theory of special creation states that life was created by.....	[U]	[1]
2	.....was first scientist to view living cell under microscope.	[U]	[1]
3	pH of the blood is…….	[A]	[2]
4	Buffers resist change in.....when small amount of acid or base are added	[K]	[2]
5	..... membrane proteins embedded within the phospholipid bilayer	[U]	[3]
6	.....transport is the movement of molecules across a cell membrane without using energy.	[U]	[3]
7	Uniport is the movement of one type of molecules in.....direction.	[U]	[4]

8	A network of membranous tubules and cisternae within the cytoplasm of eukaryotic cell is called.....	[U]	[4]
9	Photolysis of water occurs in…….. photophosphorylation	[K]	[5]
10	The first stable product of C4 cycle is.....	[K]	[5]

(10 × 1 = 10)

Part B

Short Questions

Answer 10 questions

Each question carries 2 marks

11	Write short note on orgin of life.	[A]	[1]
12	State Theory of special creation .	[K]	[1]
13	State Theory of biogenesis.	[K]	[1]
14	Explain Louis Pasteur Swan neck Flask Experiment.	[U]	[1]
15	Explain the difference between strong and weak acid.	[U]	[2]
16	Describe pH scale.	[U]	[2]
17	Describe amphitropic proteins.	[U]	[3]
18	Write short note on simple diffusion.	[K]	[3]
19	Write short note on simple diffusion.	[K]	[4]
20	Define Osmosis.	[K]	[4]
21	Write short note on non cyclic photophosphorylation.	[A]	[5]
22	Explain the role of RuBisCo in Carbon fixation.	[U]	[5]

(10 × 2 = 20)

Part C

Short Essay Type Questions

Answer 5 questions

Each question carries 4 marks

23	Define biogenesis and explain its significance in understanding life's origin.	[K]	[1]
24	Discuss the Scope of biochemistry in modern biology.	[U]	[1]
25	Draw structure of water molecule with their bond angle.	[K]	[2]

26	Explain the mechanism of facilitated diffusion with diagram.	[U]	[3]
27	Describe the process of Osmosis.	[U]	[4]
28	Contrast between symbiotic and non symbiotic nitrogen fixation.	[An]	[5]
29	Write an essay onlarger subunit of nitrogense enzyme.	[A]	[5]

$(5 \times 4 = 20)$

END OF THE QUESTION PAPER

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